

TRAINED AGENTIC CONTEXT MANAGEMENT

Anonymous authors

Paper under double-blind review

ABSTRACT

We study long context language models. Instead of training long context natively, or designing a long context harness, we train a model over the simplest possible harness: a tool to call itself with any specified prompt and a tool to read tokens in a range from the input context. We finetune Qwen3.6-35B-A3B over this harness. With only 8,000 tokens of context, our small model outperforms GPT-5.4 with 1M tokens of context on the OOLONG-synth benchmark when document length exceeds 40K tokens.

1 INTRODUCTION

As language models boast impressive breakthroughs (Luong & Lockhart, 2025; Wei, 2025; OpenAI, 2025; Lin & Cheng, 2025; OpenAI, 2026c;b), increasing focus is placed on their limitations. The most salient resource constraint of a language model is its context length. Models can process and produce tokens within this limit, but the limit is hard. Model producers have identified this constraint and placed great emphasis on extending it as large as possible, with some model builders touting 10M (Dangel, 2026; Meta AI, 2025; Zhu et al., 2026) and even 100M (Magic, 2024) token context lengths (for reference, the length of the first Harry Potter book is about 100,000 tokens, the full U.S. tax code roughly 3M, and the Linux kernel source code about 300M).

While native context scaling is appealing as a concrete engineering problem, it is not the only approach to producing a model capable of processing long documents. In 2021, OpenAI fine-tuned GPT-3 to summarize sections of books, and then summarize those summaries, achieving competitive performance with human book summaries on a model with a context length of just 2,048 tokens (Wu et al., 2021). As the field has matured, agentic harnesses have become viable. MemGPT treats context similarly to an operating system, relying on LLM-invoked function calls to push relevant information in and out of the context window to external memory, effectively achieving expanded virtual context (Packer et al., 2023). More recently, Recursive Language Models introduced a harness that utilizes LLM’s coding prior to call itself in a REPL environment, achieving stronger performance than base models on retrieval and aggregation tasks both within context limits, but also beyond it, expanding effective context by $\sim 4\times$ (Zhang et al., 2025).

While handcrafted prompts and programmatic environments can be effective in solving long context problems without native long context training, it begs an obvious question: if models can, for example, solve all the problems at the International Collegiate Programming Contest¹ World Finals (OpenAI, 2025), or more generally, implement requested features in codebases far larger than context limits (Marten et al., 2026; OpenAI, 2026a; Anthropic, 2026), can they not be *trained* to manage context effectively, *without human-authored harness crutches*? Such model training internalizes decisions external harnesses hardcode, offering the potential to scale much more generally.

We answer this question in this paper, affirmatively. We consider the simplest possible agentic context harness, defined below.

1.1 HARNESS

Our harness has two tools:

1. `read_chunk(start, end)`: Read a slice of the document and return the decoded text of tokens `[start, end)`.

¹<https://icpc.global/>

2. `spawn_subagent(subtask)`: Spawn a fresh-context copy of yourself to solve `subtask`; returns the child’s `\boxed{}` answer.

Unmodified long-context prompts are provided to the model over a document accessed via the above harness. The system prompt is propagated to child agents and includes the above harness definition as well as the original prompt²; child agents’ user prompt is specified by the calling model.

1.2 RESULTS

We finetune Qwen3.6-35B-A3B on the above harness and achieve the following results:

1. blah
2. blah blah

1.3 PAPER ORGANIZATION

In the next section, we frame our work amongst others. We believe related work belongs at the *start* of a paper, where it is most useful in articulating where our contributions fit in the literature. In Section 3, we describe our training methodology. In Section 4, we discuss all our experiments, and we conclude in Section 5.

2 RELATED WORK

Long-context LLM work can be broken into a few distinct categories: *native long context models*, *context compaction*, and *long context harnesses*, both trained and untrained. We discuss the first two briefly and the last at greater length, as it is most relevant to this work.

2.1 NATIVE LONG CONTEXT MODELS

3 METHODOLOGY

4 EXPERIMENTS

5 CONCLUSION

AI USE STATEMENT

In this work, we used generative AI tools to write all our code, administer experiments, create figures, search and analyze relevant literature, format references, generate synthetic data sets, propose or refine hypotheses/brainstorming, design or provide feedback on research methodology/experiments, clean and reformat data, support qualitative and thematic data analysis, and interpret results. We have not used generative AI tools to draft parts of this research paper, and the following topics are not applicable to this work: help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical claims, assist in the writing of proofs, and assist with translation. We have reviewed our AI-assisted work, for example, by reviewing AI code and conducting manual audits of synthetic data. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REFERENCES

Anthropic. Introducing Claude Fable 5.1 and Claude Mythos 5.1. Anthropic blog post, September 2026. URL <https://www.anthropic.com/claude-fable-and-mythos-5-1>.

²The full system prompt and harness environment is elucidated in Appendix A.

Justin Dangel. Introducing SubQ: The first fully subquadratic LLM. Subquadratic blog post, May 2026. URL <https://subq.ai/introducing-subq>. 12M-token research result; the shipped model is SubQ 1M-Preview.

Hanzhao Lin and Heng-Tze Cheng. Gemini achieves gold-medal level at the International Collegiate Programming Contest World Finals. Google DeepMind blog post, September 2025. URL <https://deepmind.google/blog/gemini-achieves-gold-medal-level-at-the-international-collegiate-programming-contest>. 10 of 12 problems; would rank 2nd among university teams.

Thang Luong and Edward Lockhart. Advanced version of Gemini with Deep Think officially achieves gold-medal standard at the International Mathematical Olympiad. Google DeepMind blog post, July 2025. URL <https://deepmind.google/blog/advanced-version-of-gemini-with-deep-think-officially-achieves-gold-medal-standard>. 35/42, five of six problems, graded by IMO coordinators.

Magic. 100M token context windows. Blog post, August 2024. URL <https://magic.dev/blog/100m-token-context-windows>. LTM-2-mini; introduces the HashHop evaluation.

Ryan Marten, Alex Shaw, Ivan Bercovich, Benedikt Droste, Tommaso Cerruti, Steven Dillmann, Mike Merrill, Andy Konwinski, and Ludwig Schmidt. Terminal-Bench: A benchmark for AI agents in terminal environments. Harbor Framework and Laude Institute, August 2026. URL <https://www.tbench.ai/>. Version 4.0.0, released 2026-08-26. Zenodo DOI 10.5281/zenodo.22105680.

Meta AI. The Llama 4 herd: The beginning of a new era of natively multi-modal AI innovation. Blog post, April 2025. URL <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>. Llama 4 Scout, 10M-token context; model card at <https://huggingface.co/meta-llama/Llama-4-Scout-17B-16E-Instruct>.

OpenAI. Our general-purpose reasoning models solved all 12 problems at the 2025 ICPC world finals. OpenAI announcement (X), September 17, 2025, September 2025. URL <https://x.com/OpenAI/status/1968368133024231902>. 12/12 under the five-hour contest rules; the best human team solved 11.

OpenAI. Introducing GPT-6 Astra. OpenAI blog post, September 2026a. URL <https://openai.com/index/gpt-6-astra/>.

OpenAI. Finite time blowup for Navier–Stokes. Manuscript, 166 pages, with Lean formalization; announced September 8, 2026, September 2026b. URL <https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf>. Forced finite-time blow-up with bounded kinetic energy (alternative (C) of the Clay problem statement); announcement at <https://openai.com/index/navier-stokes-solution/>.

OpenAI. Ten advances in mathematics and theoretical computer science. OpenAI blog post, August 1, 2026, August 2026c. URL <https://openai.com/index/ten-advances-in-mathematics/>. Astra; ten problems open for ten or more years, each with a Lean 4 proof certificate.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems, 2023. URL <https://arxiv.org/abs/2310.08560>.

Alexander Wei. An experimental reasoning LLM achieves gold medal-level performance on the International Math Olympiad. OpenAI announcement (X thread), July 19, 2025, July 2025. URL https://x.com/alexwei_/status/1946477742855532918. 35/42 under contest conditions; proofs graded by three former IMO medalists.

Jeff Wu, Long Ouyang, Daniel M. Ziegler, Nisan Stiennon, Ryan Lowe, Jan Leike, and Paul Christiano. Recursively summarizing books with human feedback, 2021. URL <https://arxiv.org/abs/2109.10862>.

Alex L. Zhang, Tim Kraska, and Omar Khattab. Recursive language models, 2025. URL <https://arxiv.org/abs/2512.24601>. v1 December 2025; revised May 2026 with the SFT-trained RLM-Qwen3-8B.

Zheqing Zhu, Christopher Wu, Yi Wan, Yiqing Cai, Qiwei He, Ruihao Zhu, Chang Zen Tee, Lexin Chen, Pu Thavikulwat, and Jared Ee. Pokee-Isaac 28B: A 10M-token context efficient agentic model. Technical report, Pokee AI, August 2026. URL <https://console.pokee.ai/pokee-isaac-28b-v0-technical-report.pdf>.

A HARNESS AND PROMPTS

You may include other additional sections here.